

# The Planck gap as a theorem: every marking protocol needs $F^2\tau^3 > 9m\kappa$

Let  $\kappa$  be the smallest product of resolution and recoil width over the marks a record model can make. Then for **every** finite protocol of marks, of any number, at any times, the falling parabola can be told from the inertial line over a duration  $\tau$  in the worst case only if

$$F^2\tau^3 > 9m\kappa, \quad \text{equivalently} \quad \tau\Delta E = \frac{F^2\tau^3}{2m} > \frac{9}{2}\kappa, \quad A_{\text{inertial,fall}} = \frac{vF\tau^3}{6m} > \frac{3v\kappa}{F},$$

and a two-mark protocol decides as soon as  $F^2\tau^3 > 36m\kappa$  when marks saturating  $\kappa$  are available (Theorems 1–2). The threshold duration is therefore pinned within a factor  $4^{1/3}$  by  $\kappa$  alone, and the Planck gap of the recorded Galileo comparison is positive exactly when  $\kappa > 0$  (Corollary 3). The same bound governs Democritus insertion: the marks inside a window of duration  $\tau'$ , used by themselves, record the force only if  $F^2\tau'^3 > 9m\kappa$  (Corollary 4). Any record model in which a probe's later position records fix its momentum to arbitrary accuracy has  $\kappa = 0$  (Proposition 5), and Newton's optics gives the heuristic value  $\kappa = \Lambda p = h/2$  of the mark-floor note. **Quantum kinematics has no non-vacuous instance of this framework:** no mark has both a finite worst-case resolution and a finite worst-case recoil width, by Paley–Wiener, so  $\kappa = \infty$  and the theorem holds emptily (Proposition 6). What stands is the classical half of the programme's question, and the quantum half is carried out in the probabilistic note, which proves  $F\tau(L + \tau P/2m) \geq (1 - 2\epsilon)\hbar$  for every protocol of quantum instruments,  $(L, P)$  being the apparatus aperture, hence  $\tau\Delta E \geq (1 - 2\epsilon)^2\hbar^2/(4LP)$  at the balanced aperture  $L = \tau P/2m$  and  $(1 - 2\epsilon)^2\hbar/2$  when that aperture has the smallest area the uncertainty relation allows. The proof is a separation argument for the adversary's feasibility set, an AM–GM step that turns  $\delta\Delta \geq \kappa$  into a square root, and one integration by parts that converts the certificate into a Dirichlet energy. Constants are explicit throughout; nothing is fitted.

## 1. Setting and definitions

Transverse coordinate  $y$ , mass  $m > 0$ , comparison on  $[0, \tau]$  between the inertial hypothesis I and the falling hypothesis F, which differ by the displacement

$$P(t) = \frac{Ft^2}{2m}, \quad P(0) = P'(0) = 0, \quad P'' = \frac{F}{m}.$$

A **mark** at time  $t_j$  has a **resolution**  $\delta_j > 0$ : the datum it leaves is an interval of length  $\delta_j$  containing  $y(t_j)$ ; and a **recoil width**  $\Delta_j \geq 0$ : the transverse impulse it delivers to the body lies, under either hypothesis, in an interval of length  $\Delta_j$  that is fixed by the totality of records, so that two runs which agree in every record can differ in that impulse by at most  $\Delta_j$ . Between marks the body moves freely under I and with the force  $F$  under F. A **protocol** is a finite set of marks at times  $0 = t_0 < t_1 < \dots < t_k \leq \tau$ ; the mark at  $t_0 = 0$  is the preparation, whose recoil width  $\Delta_0$  is the width within which the records fix the initial transverse momentum. A protocol **decides** the comparison in the worst case when no pair of runs, one under I and one under F, with initial positions, initial momenta and recoils allowed by the marks, produces the same data. Set

$$\kappa = \inf_j \delta_j \Delta_j$$

over the marks the model can make; a model satisfies the **mark trade-off** with constant  $\kappa$  when every available mark has  $\delta\Delta \geq \kappa$ . For the corpuscular model of the mark-floor note this is premise M3 with  $\kappa = \Lambda p$ .

**Feasibility system.** Write  $y_F - y_I = P + d$ . Then  $d(t) = x + vt + \sum_{j \geq 1} \rho_j(t - t_j)_+/m$  with  $x \in \mathbb{R}$  free (both initial positions lie in the same datum interval, so their difference is bounded only by the datum constraint at  $t_0$ ),  $|v| \leq \Delta_0/m$  and  $|\rho_j| \leq \Delta_j$ . Two runs give the same data exactly when

$$|d(t_i) + P(t_i)| \leq \delta_i \quad (i = 0, 1, \dots, k). \quad (1)$$

The protocol decides if and only if (1) has no solution  $(x, v, \rho_1, \dots, \rho_{k-1})$  in the stated ranges. (The last mark's recoil affects nothing that is recorded and is omitted.)

## 2. Certificate of decidability

**Lemma 1 (separation).** The protocol decides if and only if there are multipliers  $\mu_0, \dots, \mu_k \in \mathbb{R}$  with  $\sum_i \mu_i = 0$  and

$$\sum_i \mu_i P(t_i) > \sum_i |\mu_i| \delta_i + \frac{\Delta_0}{m} |S_0| + \sum_{j \geq 1} \frac{\Delta_j}{m} |S_j|, \quad S_j = \sum_{i > j} \mu_i (t_i - t_j). \quad (2)$$

*Proof.* Let  $L : (x, v, \rho) \mapsto (d(t_i))_i \in \mathbb{R}^{k+1}$  be the linear map of Section 1 and  $B$  the box  $|v| \leq \Delta_0/m, |\rho_j| \leq \Delta_j$ . The system (1) is feasible exactly when the closed convex set  $C = \{L(x, v, \rho) : x \in \mathbb{R}, (v, \rho) \in B\}$  meets the compact box  $Q = \{q : |q_i + P(t_i)| \leq \delta_i\}$ .  $C$  is the sum of a line and a compact convex set, hence closed and convex. If  $C \cap Q = \emptyset$ , the separation theorem for a closed and a compact convex set gives  $\mu$  with  $\sup_{q \in Q} \mu \cdot q < \inf_{c \in C} \mu \cdot c$ . Since  $x$  is free the infimum is  $-\infty$  unless  $\mu \cdot L(1, 0, 0) = \sum_i \mu_i = 0$ ; with that,  $\inf_C \mu \cdot c = -\frac{\Delta_0}{m} |\sum_i \mu_i t_i| - \sum_{j \geq 1} \frac{\Delta_j}{m} |\sum_{i > j} \mu_i (t_i - t_j)|$  by choosing the signs of  $v$  and  $\rho_j$ , and  $\sup_Q \mu \cdot q = -\sum_i \mu_i P(t_i) + \sum_i |\mu_i| \delta_i$ . The strict inequality between them is (2); if the separation comes with the opposite orientation, replace  $\mu$  by  $-\mu$ . Note  $\sum_i \mu_i t_i = S_0$  when  $\sum_i \mu_i = 0$ . Conversely, if (2) holds and  $(x, v, \rho)$  solved (1), pairing (1) with  $\mu$  would give  $\sum_i \mu_i P(t_i) \leq \sum_i |\mu_i| \delta_i - \sum_i \mu_i d(t_i) \leq \sum_i |\mu_i| \delta_i + \frac{\Delta_0}{m} |S_0| + \sum_{j \geq 1} \frac{\Delta_j}{m} |S_j|$ , contradicting (2).  $\square$

**Lemma 2 (trade-off to a square root).** If  $\delta_j \Delta_j \geq \kappa$  for all  $j$ , then the right side of (2) is at least  $2\sqrt{\kappa/m} \sum_{j=0}^k \sqrt{|\mu_j| |S_j|}$ .

*Proof.* For each  $j$ ,  $|\mu_j| \delta_j + \frac{\Delta_j}{m} |S_j| \geq 2\sqrt{|\mu_j| |S_j| \delta_j \Delta_j / m} \geq 2\sqrt{\kappa |\mu_j| |S_j| / m}$  by the arithmetic–geometric mean inequality;  $S_k = 0$ .  $\square$

## 3. The universal lower bound

**Lemma 3 (the certificate as a Dirichlet energy).** Let  $\sum_i \mu_i = 0$ ,  $N(u) = \sum_{i: t_i > u} \mu_i$  and  $T(u) = \int_u^\tau N(w) dw$  for  $u \in \mathbb{R}$ . Then  $T$  is continuous and piecewise linear,  $T \equiv T(0)$  on  $(-\infty, 0]$ ,  $T \equiv 0$  on  $[\tau, \infty)$ ,  $T(t_j) = S_j$ , the slope of  $T$  jumps by  $\mu_j$  at  $t_j$ , and

$$\sum_i \mu_i P(t_i) = \frac{F}{m} \int_0^\tau T(u) du, \quad \sum_j \mu_j T(t_j) = - \int_0^\tau T'(u)^2 du. \quad (3)$$

*Proof.*  $N$  is piecewise constant, vanishes for  $u < 0$  because  $\sum_i \mu_i = 0$  and for  $u \geq t_k$ , and drops by  $\mu_j$  as  $u$  crosses  $t_j$ ; so  $T' = -N$  has the stated jumps and support. Also  $T(t_j) = \sum_i \mu_i |\{u \in [t_j, \tau] : u < t_i\}| = \sum_{i > j} \mu_i (t_i - t_j) = S_j$ . For the first identity,  $P(t_i) = \int_0^{t_i} P'(u) du$  and Fubini give  $\sum_i \mu_i P(t_i) = \int_0^\tau P'(u) N(u) du = - \int_0^\tau P' T' du = -[P' T]_0^\tau + \int_0^\tau P'' T du = \frac{F}{m} \int_0^\tau T du$ , using  $T(\tau) = 0$  and  $P'(0) = 0$ . For the second,  $T'$  is of bounded variation with jumps  $\mu_j$  at  $t_j$  and no other variation, so  $\sum_j \mu_j T(t_j) = \int_{\mathbb{R}} T dT' = [TT']_{-\infty}^\infty - \int T'^2 = - \int_0^\tau T'^2$ , as  $T'$  vanishes outside  $[0, \tau]$ .  $\square$

**Theorem 1 (universal lower bound).** Suppose every mark of a protocol satisfies  $\delta_j \Delta_j \geq \kappa$ . If the protocol decides the Galileo comparison on  $[0, \tau]$  in the worst case, then

$$F^2\tau^3 > 9m\kappa.$$

*Proof.* By Lemma 1 there is  $\mu$  with  $\sum_i \mu_i = 0$  satisfying (2), and by Lemma 2 and (3)

$$\frac{F}{m} \int_0^\tau T du > 2\sqrt{\frac{\kappa}{m}} \sum_j \sqrt{|\mu_j| |S_j|} \geq 2\sqrt{\frac{\kappa}{m}} \sqrt{\sum_j |\mu_j| |S_j|} \geq 2\sqrt{\frac{\kappa}{m}} \sqrt{E}, \quad E = \int_0^\tau T'^2 du,$$

using  $\sum_j \sqrt{a_j} \geq \sqrt{\sum_j a_j}$  for  $a_j \geq 0$  and  $\sum_j |\mu_j| |S_j| \geq |\sum_j \mu_j T(t_j)| = E$ . The right side is nonnegative, so  $\int_0^\tau T > 0$ . Since  $T(\tau) = 0$ ,  $|T(u)| = |\int_u^\tau T'| \leq \sqrt{\tau-u} \sqrt{E}$  by Cauchy-Schwarz, hence  $\int_0^\tau T \leq \int_0^\tau \sqrt{\tau-u} du \sqrt{E} = \frac{2}{3} \tau^{3/2} \sqrt{E}$ , that is  $\sqrt{E} \geq \frac{3}{2} \tau^{-3/2} \int_0^\tau T$ . Inserting,  $\frac{F}{m} \int_0^\tau T > 3\sqrt{\kappa/m} \tau^{-3/2} \int_0^\tau T$ , and dividing by the positive integral,  $F\tau^{3/2} > 3\sqrt{\kappa m}$ .  $\square$

The theorem uses nothing about the marks except the product bound: no least length, no least impulse separately, no restriction on how many marks are made or when, no assumption that the marks are of one kind, and no probabilistic model. In the mark-floor note's two-mark protocol the same inequality appeared with the constant 16; Theorem 1 says that no protocol whatever can push the constant below 9.

#### 4. The matching upper bound and the gap

**Theorem 2 (two marks suffice).** Suppose the model offers, for the duration  $\tau$  in question, a preparation mark with  $\delta_0 = \sqrt{\kappa\tau/m}$  and  $\Delta_0 = \kappa/\delta_0$ , and a terminal mark with  $\delta_1 \leq \sqrt{\kappa\tau/m}$ . Then the two-mark protocol decides the comparison whenever  $F^2\tau^3 > 36m\kappa$ .

*Proof.* Under I the terminal position lies in an interval of length  $w = \delta_0 + \Delta_0\tau/m = 2\sqrt{\kappa\tau/m}$ ; under F in its translate by  $s(\tau) = F\tau^2/2m$ . A datum of length  $\delta_1$  cannot meet both when  $s(\tau) > w + \delta_1$ , which holds if  $F\tau^2/2m > 3\sqrt{\kappa\tau/m}$ , that is  $F^2\tau^3 > 36m\kappa$ .  $\square$

**Corollary 3 (the Planck gap of the recorded comparison).** Let  $\tau_*(F, m)$  be the infimum of durations over which some available protocol decides the comparison. Under the hypotheses of Theorems 1–2,

$$9m\kappa \leq F^2\tau_*^3 \leq 36m\kappa, \quad \frac{9}{2}\kappa \leq \tau_*\Delta E(\tau_*) \leq 18\kappa, \quad \frac{3v\kappa}{F} \leq A(\tau_*) \leq \frac{12v\kappa}{F},$$

the lower bounds from Theorem 1 as infima and the upper bounds from Theorem 2.

In particular the recorded comparison has a positive floor if and only if  $\kappa > 0$ , and the floor is a fixed multiple of  $\kappa$ , between  $\frac{9}{2}$  and 18 in the action-scaled variable  $\tau\Delta E$ . The geometric area inherits the factor  $v/F$  and has no floor of its own, which is the Q14 gate: the only action the record model contains is  $\kappa$ .

**Corollary 4 (Democritus insertion).** Let marks be inserted at  $t_a = t_j < t_{j+1} < \dots < t_{j+r} = t_b$  inside  $[0, \tau]$ , and let the observer use these marks alone to decide whether the motion on  $[t_a, t_b]$  was free or forced, with no information about the momentum at  $t_a$  beyond what these marks give. Then the window records the force only if  $F^2(t_b - t_a)^3 > 9m\kappa$ .

*Proof.* Shift the origin to  $t_a$ . With the momentum at  $t_a$  unknown,  $v$  is free as well as  $x$ , so the certificate of Lemma 1 acquires the extra condition  $S_0 = \sum_i \mu_i t_i = 0$  and loses the term  $\frac{\Delta_0}{m} |S_0|$ , which was zero under that condition anyway. Lemmas 2–3 and the proof of Theorem 1 go through unchanged with  $\tau$  replaced by  $t_b - t_a$ .  $\square$

So inserted points finer than  $\tau_* = (9m\kappa/F^2)^{1/3}$  show, by themselves, free motion only; the force reappears when enough of them are pooled that the pooled window exceeds  $\tau_*$ , and it also reappears if momentum information from before the window is carried in, in which case the comparison is the whole-interval one and Theorem 1 applies to it. This is the rigorous form of the mark-floor note's Theorem 2, and of I003's remark 3:  $\kappa$  controls the mesh at which the marked polygon stops converging to Newton's curve.

## 5. What fixes $\kappa$

**Proposition 5 (deterministic probes give  $\kappa = 0$ ).** Suppose the marks are made by probes that travel freely after the mark, that a probe’s transverse momentum after the mark equals its momentum before minus the impulse delivered, and that a probe’s position can be recorded at any distance  $D$  with a resolution  $\delta_s$  bounded independently of  $D$ . Then for every mark  $\Delta \leq p_{\parallel}(\delta + \delta_s)/D$  for all  $D$ , where  $p_{\parallel}$  is the probe’s longitudinal momentum, so  $\Delta = 0$  and  $\kappa = 0$ .

*Proof.* The probe’s transverse momentum after the mark is  $p_{\parallel}$  times the tangent of its direction, which the two recorded positions fix to within  $(\delta + \delta_s)/D$ ; momentum conservation transfers the same width to the delivered impulse; let  $D \rightarrow \infty$ .  $\square$

This is the far-screen countermodel of the mark-floor note stated for any deterministic probe, and it is the mechanism behind the classical closures C068–C123 (synthesis): full-state records fix the recoil, the trade-off constant is zero, and Corollary 3 gives floor zero. A positive  $\kappa$  therefore requires that the probe’s later position records **do not** fix its momentum, which is an indeterminacy of the probe’s own kinematics.

**Proposition 6 (the worst-case framework has no quantum instance).** In quantum kinematics no mark has both a finite worst-case resolution and a finite worst-case recoil width. Hence  $\kappa = \infty$ , Theorem 1 holds emptily, and no protocol of marks decides the comparison with worst-case certainty.

*Proof.* Suppose a mark’s datum confines the position to an interval  $I$  of length  $\delta < \infty$  with certainty, so that every conditional post-mark state  $\psi$  has  $\text{supp } \psi \subseteq I$ . By the Paley–Wiener theorem  $\hat{\psi}$  extends to an entire function of exponential type; a nonzero entire function cannot vanish on a set with nonempty interior, so the momentum distribution  $|\hat{\psi}|^2$  has full support. The impulse the mark delivered is the difference between the post-mark and pre-mark momenta, so no bounded interval fixed by the records contains it with certainty, and  $\Delta = \infty$ . Symmetrically, a mark with  $\Delta < \infty$  confines momentum to a bounded set with certainty and so has  $\delta = \infty$ .  $\square$

This is the decisive limitation of the present theorem, and it is structural rather than technical. Worst-case interval widths express a record model in which a datum *excludes* values outright. Quantum data exclude nothing: every position outcome leaves momentum tails, and every finite sequence of finite-precision quantum measurements leaves the two hypotheses compatible with the data at some probability. So Theorem 1, Theorem 2 and Corollaries 3–4 are theorems about **classical bounded-error record models**, with Proposition 5 supplying the mechanism that sets  $\kappa = 0$  there. The quantum statement of the same combination is Corollary 3 of the probabilistic note, proved by a different route. **Exactly what was wrong with the first version of this section.** Its Proposition 6 asserted  $\kappa \geq 2\hbar$  for quantum marks and drew the floor  $\tau\Delta E > 9\hbar$ . Both statements are true, and both are empty:  $\kappa = \infty$  satisfies  $\kappa \geq 2\hbar$ , and Theorem 1’s hypothesis is then unsatisfiable, so its conclusion holds vacuously. The defect was the proof and the reading rather than the statements. The proof bounded the Busch–Lahti–Werner calibration disturbance by half a worst-case momentum width; that disturbance is a Wasserstein-2 distance, which needs a finite second moment, and the post-mark momentum distribution after a sharp position confinement has tails too heavy for one. So the argument read a finite bound off a quantity that is infinite, and the reading presented an empty inequality as a substantive quantum floor. The repair is the theorem of the probabilistic note, which reaches a genuine quantum floor of the same combination by abandoning worst-case widths.

**What the probabilistic version is.** It is carried out in the companion note, which reaches the floor  $F\tau(L + \tau P/2m) \geq (1 - 2\epsilon)\hbar$  at aperture  $(L, P)$  without any error–disturbance relation, using Mandelstam–Tamm and the uncertainty relation alone. The rest of this section records the requirement it had to meet. Replace the decision criterion by a hypothesis test: two prepared ensembles, a fixed error probability  $\epsilon$ , and a criterion that the data distinguish I from F at level  $\epsilon$ . Replace  $\delta$  and  $\Delta$  by the error and disturbance of an instrument in a stated metric. The expected shape of the conclusion,  $F^2\tau^3 \gtrsim m\hbar$ , is the standard quantum limit for detecting a force on a free

mass (Braginsky and Khalili, *Quantum Measurement*, 1992, metadata). That limit is contested in exactly the regime this note needs: Yuen argued it can be beaten, Caves defended it (PRL **54**, 2465, 1985, abstract), and Ozawa exhibited a measurement breaking the standard quantum limit for free-mass position (PRL **60**, 385, 1988, abstract) using an instrument whose disturbance is correlated with its outcome rather than independent of it. The same distinction reappears in the error–disturbance literature, where Ozawa’s counterexamples to the naive Heisenberg product coexist with the proved calibration-based relation of Busch, Lahti and Werner (PRL **111**, 160405, 2013, abstract; J. Math. Phys. **55**, 042111, 2014, abstract). A probabilistic version of Theorem 1 is therefore not a corollary of any single error–disturbance inequality. It has to say which instruments are admitted, and its interest lies precisely in whether the protocol–universality proved here (over all finite numbers of marks at all times) survives when Ozawa’s outcome-correlated disturbances are admitted.

**The Newton-age value.** The corpuscular premises M1–M3 of the mark-floor note give  $\kappa = \Lambda p$ , with  $\Lambda = 1/89000$  inch measured by Newton and  $p$  unmeasured in his age; on the identification  $p = h/\lambda$ ,  $\Lambda = \lambda/2$  this is  $\kappa = h/2$ . Newton-age materials give Corollary 3 in full once M3 is granted, and Proposition 5 shows that Newton’s own determinate fits deny M3 and give  $\kappa = 0$ . The corpuscular model is a classical bounded-error record model, which is why it fits this framework and why its value of  $\kappa$  is a heuristic analogy rather than a quantum result. The logical structure established here is therefore:

$$\begin{aligned} \text{floor of the classically recorded comparison} &> 0 && \iff && \kappa > 0, \\ \text{deterministic probes} &\implies && \kappa = 0, \end{aligned}$$

proved here for all finite mark protocols with explicit constants. No step assumes a path–integral phase rule, a fixed measurement budget, or a supplied action unit other than  $\kappa$  itself, and  $\kappa$  is characterized rather than supplied on the classical side. The quantum side is open: Proposition 6 shows this framework cannot reach it, and names the reformulation that could.

## 6. Consequence for STATE

What is proved: Theorems 1–2 sandwich the threshold of the recorded Galileo comparison between  $9m\kappa$  and  $36m\kappa$  in  $F^2\tau^3$  for every finite protocol of marks with  $\delta\Delta \geq \kappa$ ; Corollary 4 does the same for a window of inserted marks used by itself; Proposition 5 gives  $\kappa = 0$  for every deterministic probe with far-field position records. Together these characterize the floor of a classical bounded-error record model by one number and identify what kills it.

What is not proved, and was wrongly claimed in the first version of this note: any quantum value of  $\kappa$ . Proposition 6 shows the worst-case framework has no non-vacuous quantum instance, so the route to  $\hbar$  runs through a probabilistic reformulation, which lands in the disputed territory of the standard quantum limit for free-mass measurement (Yuen; Caves 1985; Ozawa 1988) and of error–disturbance relations (Ozawa; Busch–Lahti–Werner 2013).

Open, in the order that matters:

1. **Done: the probabilistic theorem.** The probabilistic note proves it for every finite adaptive protocol and decides the universality question: universality over instruments holds, universality over preparations fails, and the resource that prices the failure is the apparatus aperture.
2. **The sharp constant.** The interval  $[9, 36]$  in  $F^2\tau^3/(m\kappa)$  is not closed; find the optimal protocol and the matching certificate.
3. **General force law.** Extend from  $P = Ft^2/2m$  to general  $P$  with  $P(0) = P'(0) = 0$ ; the proof already pairs  $\int_0^\tau P''T$  against the Dirichlet energy of  $T$ , so the natural statement bounds a norm of  $P''$  from below.
4. **Why  $\kappa > 0$  without quantum kinematics.** Proposition 5 names exactly what must fail: far-field position records fixing a probe’s momentum.